Find Largest Element in array:

```
import java.util.Scanner;


public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);


        int n = sc.nextInt();
        int[] arr = new int[n];


        for (int i = 0; i < n; i++) {
            arr[i] = sc.nextInt();
        }


        int largest = arr[0];


        for (int i = 1; i < n; i++) {
            if (arr[i] > largest) {
                largest = arr[i];
            }
        }


        System.out.println(largest);
    }
}
```

Find Smallest Element In Array:

```
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        int n = sc.nextInt();
        int[] arr = new int[n];

        for (int i = 0; i < n; i++) {
            arr[i] = sc.nextInt();
        }
        int smallest = arr[0];

        for (int i = 1; i < n; i++) {
            if (arr[i] < smallest) {
                smallest = arr[i];
            }
        }

        System.out.println(smallest);

        sc.close();
    }
}
```

Find First Element In array:

```
import java.util.Scanner;


public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);


        int n = sc.nextInt();
        int[] arr = new int[n];


        for (int i = 0; i < n; i++) {
            arr[i] = sc.nextInt();
        }


        System.out.println(arr[0]);


        sc.close();
    }
}
```

Find Last Element in array

```
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        int n = sc.nextInt();
        int[] arr = new int[n];

        for (int i = 0; i < n; i++) {
            arr[i] = sc.nextInt();
        }

        System.out.println(arr[n - 1]);

        sc.close();
    }
}
```

Largest Element in 2 array:

```
import java.util.Scanner;


public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);


        int n = sc.nextInt();


        int[] arr1 = new int[n];
        int[] arr2 = new int[n];
        int[] result = new int[n];


        for (int i = 0; i < n; i++) {
            arr1[i] = sc.nextInt();
        }


        for (int i = 0; i < n; i++) {
            arr2[i] = sc.nextInt();
        }


        for (int i = 0; i < n; i++) {
            result[i] = arr1[i] + arr2[i];
        }
```

```
        int largest = result[0];

        for (int i = 1; i < n; i++) {
            if (result[i] > largest) {
                largest = result[i];
            }
        }

        System.out.println(largest);

        sc.close();
    }
}
```

Reverse an array:

Type 1:

```
import java.util.Scanner;


public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);


        int n = sc.nextInt();
        int[] arr = new int[n];


        for (int i = 0; i < n; i++) {
            arr[i] = sc.nextInt();
        }


        for (int i = n - 1; i >= 0; i--) {
            System.out.print(arr[i] + " ");
        }


        sc.close();
    }
}
```

Type 2:

```
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        int n = sc.nextInt();
        int[] arr = new int[n];

        for (int i = 0; i < n; i++) {
            arr[i] = sc.nextInt();
        }

        int start = 0;
        int end = n - 1;

        while (start < end) {
            int temp = arr[start];
            arr[start] = arr[end];
            arr[end] = temp;

            start++;
            end--;
        }
```

```
        for (int num : arr) {
            System.out.print(num + " ");
        }

        sc.close();
    }
}
```

Linear search:

```
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        int n = sc.nextInt();
        int[] arr = new int[n];

        for (int i = 0; i < n; i++) {
            arr[i] = sc.nextInt();
        }

        int key = sc.nextInt();

        int index = -1;
```

```
        for (int i = 0; i < n; i++) {
            if (arr[i] == key) {
                index = i;
                break;
            }
        }

        System.out.println(index);

        sc.close();
    }
}
```

Frequency of elements:

```
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        int n = sc.nextInt();
        int[] arr = new int[n];
        boolean[] visited = new boolean[n];
```

```java
        for (int i = 0; i < n; i++) {
            arr[i] = sc.nextInt();
        }

        for (int i = 0; i < n; i++) {
            if (visited[i]) {
                continue;
            }

            int count = 1;

            for (int j = i + 1; j < n; j++) {
                if (arr[i] == arr[j]) {
                    count++;
                    visited[j] = true;
                }
            }

            System.out.println(arr[i] + " -> " + count);
        }

        sc.close();
    }
}
```

Remove duplicates:

```
import java.util.Scanner;


public class Main {
    public static void main(String[] args) {


        Scanner sc = new Scanner(System.in);


        int n = sc.nextInt();
        int[] arr = new int[n];


        for(int i = 0; i < n; i++) {
            arr[i] = sc.nextInt();
        }


        for(int i = 0; i < n; i++) {


            boolean isDuplicate = false;


            for(int j = 0; j < i; j++) {
                if(arr[i] == arr[j]) {
                    isDuplicate = true;
                    break;
                }
            }
```

```
            if(!isDuplicate) {
                System.out.print(arr[i] + " ");
            }
        }

        sc.close();
    }
}
```

Program :

```
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        int n = sc.nextInt();
        int[] arr = new int[n];

        int sum = 0;

        for (int i = 0; i < n; i++) {
            arr[i] = sc.nextInt();

            if (arr[i] > 0) {
                sum += arr[i];
```

```
            }
        }

        System.out.println("The sum of the positive numbers in the array is " + sum);

        sc.close();
    }
}
```

Program 8:

```
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        int n = sc.nextInt();
        int[] arr = new int[n];

        int sum = 0;

        for (int i = 0; i < n; i++) {
            arr[i] = sc.nextInt();

            if (arr[i] % 2 == 0) {
```

```
            sum += arr[i];
        }
    }

    System.out.println("The sum of the even numbers in the array is " + sum);

    sc.close();
  }
}
```

Program 9:

```
import java.util.Scanner;

public class Main {
  public static void main(String[] args) {

    Scanner sc = new Scanner(System.in);

    int n = sc.nextInt();

    int[] arr1 = new int[n];
    int[] arr2 = new int[n];

    for (int i = 0; i < n; i++) {
      arr1[i] = sc.nextInt();
    }
```

```
        for (int i = 0; i < n; i++) {
            arr2[i] = sc.nextInt();
        }

        boolean compatible = true;

        for (int i = 0; i < n; i++) {
            if (arr1[i] < arr2[i]) {
                compatible = false;
                break;
            }
        }

        if (compatible) {
            System.out.println("Compatible");
        } else {
            System.out.println("Incompatible");
        }

        sc.close();
    }
}
```